

Arctic Network

There are P outposts located in the Arctic. You want to build a communication network using radio transceivers and satellite phones.

Every outpost will be equipped with a radio transceiver, but you can only install satellite phones at up to S outposts. All transceivers will have the same operation range, D , so a pair of outposts can communicate directly via radio if the Euclidean distance between them is at most D . Additionally, any pair of outposts equipped with satellite phones can communicate directly, over any distance. Outposts can also communicate indirectly through any sequence of transceivers and satellite phones.

You are given S , P , and an array `outposts` containing the 2D coordinates of all P outposts. Compute the square of the minimum transceiver range that allows the S satellite phones to connect the entire network.

Example Test Cases:

Test Case 1

Input: $S = 2$, $P = 3$, `outposts` = [(0,0), (0,4), (3,4)]

Output: 9

Explanation:

Install satellite phones at the first and second outposts. The outpost at (3,4) can communicate with the one at (0,4) via radio transceiver with a range of 3. The square of the minimum range to connect the network is 9.

Test Case 2

Input: $S = 1$, $P = 3$, `outposts` = [(0,0), (0,4), (3,4)]

Output: 16

Explanation:

There is only one satellite phone available, so all communication is by radio. Connecting (3,4) to (0,4) requires a range of 3, and connecting (0,0) to (0,4) requires a range of 4. The square of the minimum range to connect the network is 16.

Test Case 3

Input: $S = 2$, $P = 4$, `outposts` = [(0,10), (0,0), (10,0), (10,10)]

Output: 100

Explanation:

Install satellite phones at the first and third outposts. The outpost at (0,0) can communicate with the one at (10,0) via radio transceiver with a range of 10, as can the outposts at (10,10) and (0,10). The square of the minimum range to connect the network is 100.

Expected Time Complexity: $O(P^2 \log P)$

Input Format:

An integer S , an integer P , and an array `outposts` of length P . Each element of `outposts` is the 2D coordinates of an outpost.

Output Format:

Return a single integer, the square of the minimum transceiver range. That is, the minimum possible range for which there is a way to install the S satellite phones that results in every pair of outposts being connected (directly or indirectly).

Constraints:

1. $1 \leq S \leq 100$
2. $S < P \leq 500$
3. $0 \leq x, y \leq 10^4$ for all (x, y) in `outposts`.
4. All elements in `outposts` are distinct.

Instructions:

You have been provided with two template files, containing the `main` and `solve` functions respectively. The `main` function, which has been completed for you, parses the input for each test case. For each test case, it calls the `solve` function, which must compute the minimum squared range.

The `solve` function must compute this in $O(P^2 \log P)$ time.

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)

Library Usage

For this lab, you are permitted to use any modules which are part of the standard library of either C++ or python. For C++, do not add any extra `#include` directives; the provided `bits/stdc++.h` covers all allowed modules.